

Project Idea

AI-Based Automated Visual Inspection System for Industrial Component Defect Detection

This project aims to develop a computer vision-based quality inspection system capable of automatically detecting defects in industrial components. The system will learn the characteristics of defect-free products and identify anomalies in new images, enabling automated quality control with minimal human intervention. By detecting and localizing defects, the proposed solution can help improve inspection accuracy, reduce manual effort, and enhance manufacturing efficiency.

Selected Dataset Categories

MVTec AD – Metal Nut & Screw

The project will utilize the **Metal Nut** and **Screw** categories from the **MVTec Anomaly Detection (MVTec AD)** dataset. These categories contain images of normal components for training and defective components for evaluation, along with pixel-level ground truth annotations for defect localization.

The system will be trained to detect defects such as:

- Surface scratches
- Structural deformities
- Thread or surface damage
- Manufacturing imperfections
- Other visual anomalies specific to metal nuts and screws

The **Metal Nut** and **Screw** categories were selected because they represent common industrial fasteners used across automotive, aerospace, machinery, and manufacturing industries. Their inclusion allows the proposed system to demonstrate robust defect detection across multiple component types, making the solution more practical and representative of real-world automated visual inspection systems.